Using FHIR and Wikidata to exchange neighborhood geo-referenced contextual data to personalize walking interventions

Xi Yang[1,2], Pauline Lubet[1], Daniel Fernández-Álvarez[3], Andra Waagmeester[1], Martin Dijst[2,4] and Ronald Cornet[1]

[1]*Department of Medical Informatics, Reusable Health Data group Amsterdam Public Health Research Institute, Methodology & Digital Health, Amsterdam, NL*

[2] *Department of Urban Development and Mobility, Luxembourg Institute of Socio-Economic Research (LISER), Luxembourg*

[3]*WESO Research Group, University of Oviedo, Oviedo, Spain*

[4]*University of Luxembourg, Luxembourg*

Abstract

Neighborhood geo-referenced contexts (NGRCs) are defined as the physical and social opportunities available in one's residential environment or along daily travel routes. NGRCs, highlighted in behavioral change theories and time geography, influence walking behaviors; however, their use in practice is limited and inconsistent. These inconsistencies impede interdisciplinary integration and evidence development.

To address the challenge of integrating heterogeneous and consistent NGRCs for personalized behavioral interventions, this poster presents an approach that leverages FHIR as a semantic pattern and Wikidata as an ontological framework to exchange, integrate, and link NGRC data from multiple sources (e.g., OpenStreetMap, Kadaster). We illustrate this approach through a use case in the Netherlands, demonstrating how NGRC-focused walking interventions can be operationalized in practice. By structuring and harmonizing NGRCs across data sources, our approach supports a more systematic understanding and application of NGRCs to optimize walking interventions and improve adherence.

Keywords

NGRCs, neighborhood georeferenced contexts, walking intervention, behavioral intervention, sementic web

1. Introduction

Walking interventions often show limited participation, engagement, and adherence. Incorporating neighborhood geo-referenced contexts (NGRCs) to personalize the intervention may help enhance these key outcomes. NGRCs are defined as the physical and social opportunities available in one's residential environment or along daily travel routes.

Despite progress in behavioral change theories[1, 2, 3, 4, 5], time geography [6], and digital tools [7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19], current approaches provide only inconsistent, coarse, and incomplete representations of NGRCs. The limited and inconsistent use of NGRCs in digital tools and clinical practice hinder understanding of how NGRCs impacts walking interventions and health outcomes. Moreover, existing ontologies[20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30], and geographic data models also fall short in representing NGRCs comprehensively. Therefore, we propose a personalized NGRC-focused walking intervention along their daily routes using NGRC public datasets in the Netherlands. This can be integrated into FHIR social care, using Wikidata as the ontological framework to represent standardized NGRCs, thereby enabling data exchange and integration and ultimately improving health outcomes.

SWAT4HCLS 2026: Semantic Web Applications and Tools for Health Care and Life Sciences, March 23–26, 2026, Amsterdam, NL

p.xiyang@amsterdamumc.nl (X. Yang); p.j.e.lubet@amsterdamumc.nl (P. Lubet); fernandezalvdaniel@uniovi.es (D. Fernández-Álvarez); a.s.waagmeester@amsterdamumc.nl (A. Waagmeester); Martin.Dijst@liser.lu (M. Dijst); r.cornet@amsterdamumc.nl (R. Cornet)

0009-0007-4874-7831 (X. Yang); 0009-0006-1234-2915 (P. Lubet); 0000-0002-8666-7660 (D. Fernández-Álvarez); 0000-0001-9773-4008 (A. Waagmeester); 0000-0002-7666-3455 (M. Dijst); 0000-0002-1704-5980 (R. Cornet)

© 2026 Copyright for this paper by its authors. Use permitted under Creative Commons License Attribution 4.0 International (CC BY 4.0).

2. Why FHIR

Fast Healthcare Interoperability Resources (FHIR)[31] is a global standard for exchanging healthcare data across systems. Although no equivalent standard currently exists for social care data, FHIR is increasingly being adopted in social care settings to enhance interoperability and data sharing. Leveraging FHIR as a single, unified platform for both health and social care data offers clear advantages for integration, interoperability, and coordinated whole-person care delivery, as it is open, free to use, and relatively easy to implement.

3. Why Wikidata

Existing ontologies (such as SNOMED CT, LOINC, SDOHO ontology[29], SDOH ontology[26]) cover only a subset of relevant NGRCs and are difficult to extend, often resulting in ad hoc combinations of multiple ontologies. In contrast, Wikidata is openly extensible, allowing anyone to add classes, properties, and mappings to external ontologies[32]. Its resolvable IRIs further enable interoperability, ensuring that data using Wikidata items and properties can be seamlessly linked and exchanged across systems.

4. Workflow

We describe the workflow of the personalized NGRC-focused walking intervention in Figure 1.

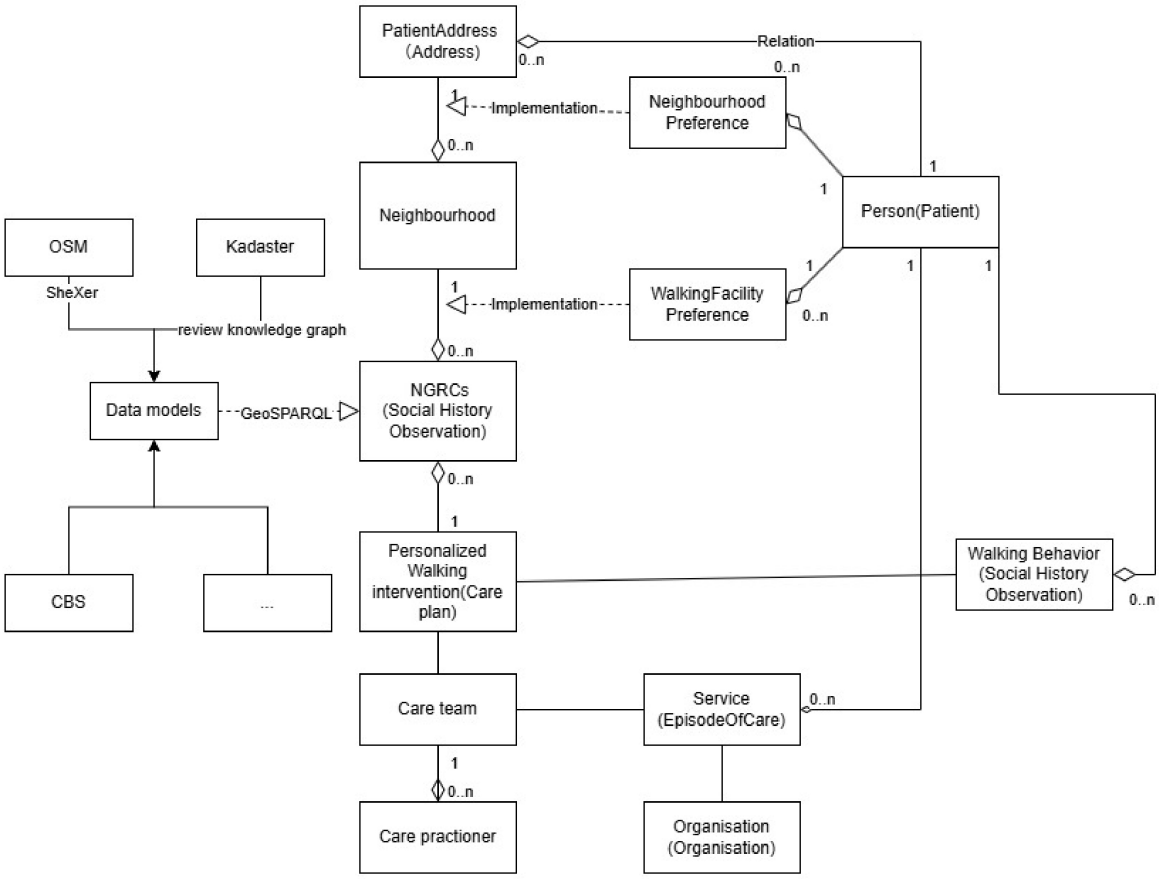

Figure 1: The workflow of personalized walking interventions using FHIR. The terms shown in parentheses represent the corresponding FHIR resource names.

5. Data model extraction of existing data sources

We use sheXer [33] to automatically extract and read ShEx schemas from data sources (e.g., OpenStreetMap (OSM) for data in the Netherlands). Also, we reviewed the data models or knowledge graph of other resources(e.g., Kadaster, Nationaal Dataportaal Wegverkeer (NDW), Centraal Bureau voor de Statistiek(CBS), maps. amsterdam) by examining their documentation (e.g., XML files and technical documents). These data models can be divided into geographic models(including vector and raster data models) and relational data models (e.g., CSV). The vector data model could be further divided. Therefore, the complexity of various data models in diverse data sources emphasizes the importance of our work.

6. Real-world scenario: Anna's use case

Anna's residential address is 1095MS, and her workplace is located at 1105AZ. She regularly goes to eat near Amsterdam Central Station. She prefers to walk in the streets with attractive architecture, sidewalk conditions, and wide sidewalks. She prefers to spend most of her time within a 500 m radius around her regular visiting places and transit stations, where she feels comfortable and enjoys walking.

Based on Anna's addresses (1095MS, 1105AZ, and 1012AB), we extract relevant neighborhood features—including sidewalk width, sidewalk maintenance, and sidewalk slope from OpenStreetMap [34], as well as architectural aesthetics from maps. amsterdam [35] —within a 500 m radius of each address.

Based on Anna's situation and personal preferences, we suggest that she walk along the streets in Amsterdam Central Station during the daytime to enjoy the colorful architecture and appealing, engaging shops, along with generally good sidewalk conditions. Although the area is crowded, the effective sidewalk width can feel somewhat limited.

Acknowledgments

The authors used AI-based tools (e.g., ChatGPT) to assist with language polishing. All content was reviewed and validated by the authors, who take full responsibility for the final manuscript.

References

[1] L. A. Basta, T. S. Richmond, D. J. Wiebe, Neighborhoods, daily activities, and measuring health risks experienced in urban environments, Social Science & Medicine 71 (2010) 1943–1950. doi:10.1016/j.socscimed.2010.09.008.

[2] K. M. Cummings, G. T. Fong, R. Borland, Environmental influences on tobacco use: Evidence from societal and community influences on tobacco use and dependence, Annual Review of Clinical Psychology 5 (2009) 433–458. doi:10.1146/annurev.clinpsy.032408.153607.

[3] M. M. Kepper, C. A. Myers, K. D. Denstel, R. F. Hunter, W. Guan, S. T. Broyles, The neighborhood social environment and physical activity: a systematic scoping review, International Journal of Behavioral Nutrition and Physical Activity 16 (2019) 124. doi:10.1186/s12966-019-0873-7.

[4] B. Popkin, K. Duffey, P. Gordon-Larsen, Environmental influences on food choice, physical activity and energy balance, Physiology & Behavior 86 (2005) 603–613. doi:10.1016/j.physbeh.2005.08.051.

[5] A. Whittal, L. Atkins, O. R. Herber, What the guide does not tell you: reflections on and lessons learned from applying the com-b behavior model for designing real life interventions, Translational Behavioral Medicine 11 (2021) 1122–1126. doi:10.1093/tbm/ibaa116.

[6] M. Dijst, Time geographic analysis, in: International Encyclopedia of Human Geography, Elsevier, 2020, pp. 271–282. doi:10.1016/B978-0-08-102295-5.10326-9.

[7] M. M. Van Stralen, H. De Vries, A. N. Mudde, C. Bolman, L. Lechner, The working mechanisms of an environmentally tailored physical activity intervention for older adults: a randomized controlled trial, International Journal of Behavioral Nutrition and Physical Activity 6 (2009) 83. doi:10.1186/1479-5868-6-83.

[8] D. A. Peels, C. Bolman, R. H. J. Golsteijn, H. De Vries, A. Mudde, M. Van Stralen, L. Lechner, Long-term efficacy of a printed or a web-based tailored physical activity intervention among older adults, International Journal of Behavioral Nutrition and Physical Activity 10 (2013) 104. doi:10.1186/1479-5868-10-104.

[9] M. N. Galea, S. R. Bray, K. A. M. Ginis, Barriers and facilitators for walking in individuals with intermittent claudication, Journal of Aging and Physical Activity 16 (2008) 69–84. doi:10.1123/japa.16.1.69.

[10] M. M. Van Stralen, H. De Vries, C. Bolman, A. N. Mudde, L. Lechner, Exploring the efficacy and moderators of two computer-tailored physical activity interventions for older adults: A randomized controlled trial, Annals of Behavioral Medicine 39 (2010) 139–150. doi:10.1007/s12160-010-9166-8.

[11] E. H. G. M. Collombon, C. A. W. Bolman, G.-J. De Bruijn, D. A. Peels, J. M. C. Van Der Velden, L. Lechner, Use and appreciation of combined computer- and mobile-based physical activity interventions within adults aged 50 years and older: Randomized controlled trial, Digital Health 10 (2024) 20552076241283359. doi:10.1177/20552076241283359.

[12] A. L. Vos, G.-J. De Bruijn, M. C. A. Klein, J. Lakerveld, S. C. Boerman, E. G. Smit, Snapp, a tailored smartphone app intervention to promote walking in adults of low socioeconomic position: Development and qualitative pilot study, JMIR Formative Research 7 (2023) e40851. doi:10.2196/40851.

[13] S. S. Martin, D. I. Feldman, R. S. Blumenthal, S. R. Jones, W. S. Post, R. A. McKibben, E. D. Michos, C. E. Ndumele, E. V. Ratchford, J. Coresh, M. J. Blaha, mactive: A randomized clinical trial of an automated mhealth intervention for physical activity promotion, Journal of the American Heart Association 4 (2015) e002239. doi:10.1161/JAHA.115.002239.

[14] K. Sporrel, S. Wang, D. F. Ettema, N. Nibbeling, B. J. A. Krose, M. Deutekom, R. D. D. De Boer, M. Simons, Just-in-time prompts for running, walking, and performing strength exercises in the built environment: 4-week randomized feasibility study, JMIR Formative Research 6 (2022) e35268. doi:10.2196/35268.

[15] C. A. Figueroa, L. Gomez-Pathak, I. Khan, J. J. Williams, C. R. Lyles, A. Aguilera, Ratings and experiences in using a mobile application to increase physical activity among university students: implications for future design, Universal Access in the Information Society 23 (2024) 821–830. doi:10.1007/s10209-022-00962-z.

[16] Y. Lin, J. Jessurun, B. de Vries, H. Timmermans, Motivate: towards context-aware recommendation mobile system for healthy living, ???? Unpublished / incomplete citation.

[17] M. Tabak, Telemedicine for patients with COPD: new treatment approaches to improve daily activity behaviour, Ph.D. thesis, University of Twente, Enschede, The Netherlands, 2014. doi:10.3990/1.9789461085900.

[18] Q. He, E. Agu, On11: an activity recommendation application to mitigate sedentary lifestyle, in: Proceedings of the 2014 Workshop on Physical Analytics, ACM, Bretton Woods, New Hampshire, USA, 2014, pp. 3–8. doi:10.1145/2611264.2611268.

[19] K. Sporrel, R. D. D. De Boer, S. Wang, N. Nibbeling, M. Simons, M. Deutekom, D. Ettema, P. C. Castro, V. Z. Dourado, B. Kröse, The design and development of a personalized leisure time physical activity application based on behavior change theories, end-user perceptions, and principles from empirical data mining, Frontiers in Public Health 8 (2021) 528472. doi:10.3389/fpubh.2020.528472.

[20] N. M. Kollapally, Y. Chen, J. Xu, J. Geller, An ontology for the social determinants of health domain, in: 2022 IEEE International Conference on Bioinformatics and Biomedicine (BIBM), IEEE, Las Vegas, NV, USA, 2022, pp. 2403–2410. doi:10.1109/BIBM55620.2022.9995544.

[21] S. Artiga, E. Hinton, Beyond health care: The role of social determinants in promoting health and health equity, https://www.kff.org/racial-equity-and-health-policy/issue-brief/

beyond-health-care-the-role-of-social-determinants-in-promoting-health-and-health-equity/, ????

[22] Kaiser Family Foundation, Medicaid authorities and options to address social determinants of health, https://www.kff.org/medicaid/medicaid-authorities-and-options-to-address-social-determinants-of-health-sdoh/, ????

[23] World Health Organization, Operational framework for monitoring social determinants of health equity, https://iris.who.int/bitstream/handle/10665/375732/9789240088320-eng.pdf, 2024.

[24] American Hospital Association, Icd-10-cm coding for social determinants of health, https://www.aha.org/system/files/2018-04/value-initiative-icd-10-code-social-determinants-of-health.pdf, ????

[25] U.S. Centers for Disease Control and Prevention, Research summary: Social determinants of health, https://www.cdc.gov/public-health-gateway/php/data-research/index.html, ????

[26] The New England Journal of Medicine Catalyst, Social determinants of health (sdoh), https://catalyst.nejm.org/doi/full/10.1056/CAT.17.0312, ????

[27] Healthy People 2030, Social determinants of health, https://odphp.health.gov/healthypeople/priority-areas/social-determinants-health/literature-summaries#block-sdohinfographics, ????

[28] HIMSS, Social determinants of health, https://legacy.himss.org/resources/social-determinants-health, ????

[29] Y. Dang, F. Li, X. Hu, V. K. Keloth, M. Zhang, S. Fu, M. F. Amith, J. W. Fan, J. Du, E. Yu, H. Liu, X. Jiang, H. Xu, C. Tao, Systematic design and data-driven evaluation of social determinants of health ontology (sdoho), Journal of the American Medical Informatics Association 30 (2023) 1465–1473.

[30] KFF, What are social determinants of health?, https://www.kff.org/medicaid/medicaid-authorities-and-options-to-address-social-determinants-of-health-sdoh/, ????

[31] Resourcelist — fhir v5.0.0, https://www.hl7.org/fhir/resourcelist.html, ???? Accessed Jan 6, 2026.

[32] A. Jacobsen, A. Waagmeester, R. Kaliyaperumal, G. S. Stupp, L. M Schriml, M. Thompson, A. I Su, M. Roos, Wikidata as an intuitive resource towards semantic data modeling in data fairification (2018).

[33] D. Fernandez-Álvarez, J. E. Labra-Gayo, D. Gayo-Avello, Automatic extraction of shapes using shexer, Knowledge-Based Systems 238 (2022) 107975.

[34] OpenStreetMap, Openstreetmap, https://www.openstreetmap.org/, ???? Accessed Jan 6, 2026.

[35] Welstand — de schoonheid van amsterdam, https://maps.amsterdam.nl/welstand/, ???? Accessed Dec 30, 2025.